



How to select solar panels for the 1.5HP sunlight pump:

Preferably, select three solar panels around 500-600 W, **with a Voc below 53V** .

Example of a suitable solar panel
(borderline case):

ELECTRICAL PARAMETERS AT STC

TYPE	JAM72D40 JAI 580/LB
Rated Maximum Power(Pmax) [W]	580
Open Circuit Voltage (Voc) [V]	51.60
Maximum Power Voltage(Vmp) [V]	43.06
Short Circuit Current(Isc) [A]	14.23
Maximum Power Current(Imp) [A]	13.47

The total Voc must be less than 160 V.



At more than 160 V
controller will be damaged



But the closer to 160 V,
the more water you will
get !

$$\text{Power (W)} = V_{mp} * I_{mp}$$

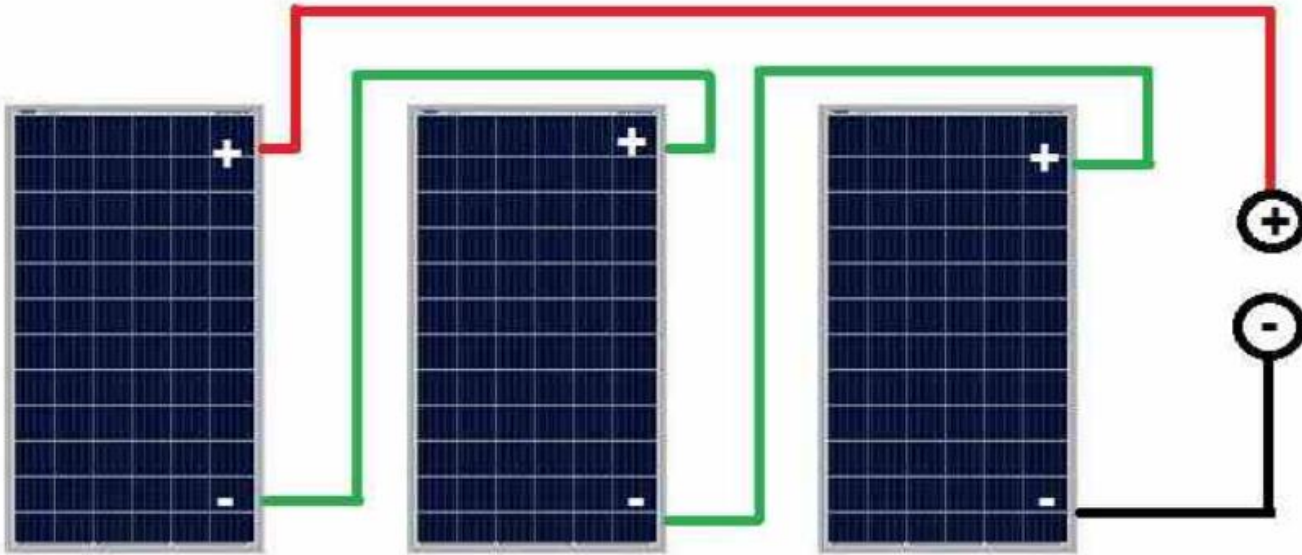
The controller limits input current at 17 A, but **PV output current can be oversized** (18 A, 19 A, etc.) to extend operating time at full discharge (e.g. during cloudy weather).



How to connect solar panels for 1.5 HP:

Connect 3 solar panels in series (+ to -, positive to negative) to obtain sufficient voltage to pump water.

In series, voltage (V) is added. In parallel, current (A) is added.



Use also a 18-35 A, DC electrical breaker (MCB) between solar panel and pump to avoid connecting cables with voltage that could lead to short sparks, and ultimately damage the controller.



Always place the MCB next to the solar panels to minimize the risk of overvoltage, and limit cable distance to 15m if you are close to the voltage limit (between 150 and 160 V).



How to place the solar panel:

Towards the **south**, if you live in the **northern hemisphere**

Towards the **north**, if you live in the **southern hemisphere**

The **angle** of the solar panel should be **equal to the latitude** of your location if you are going to use the pump all year round.



Avoid shadows on the solar modules

Use longer cables to move the solar panel away from disturbing vegetation

Clean the solar modules regularly from dirt

How to make hose connections:

Use **silicone sealant** to seal pipe fittings to optimize pump performance. Pump performance will be reduced if it draws in air.



Use **hot oil** to soften the pipe, before inserting the connector in the pipe.

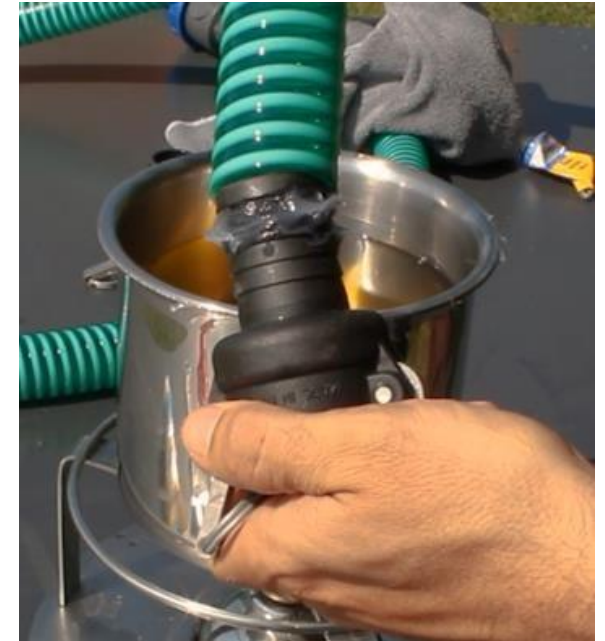


Do not use a fire flame to soften, it will damage the pipe



Use enough **hose clamps** to tighten the connections so that no air/water leaks occur.

Check this video tutorial: [TIP 1: Airproof connections](#)



How to prepare suction side:

Only use 1 ½" suction hose reinforced with PVC rings.

Do not use poor quality suction hose, or any other type of hose, as they may become damaged over time and the pump will suck in air.

Do not use multiple pieces, as each connection could cause the pump to draw in air.

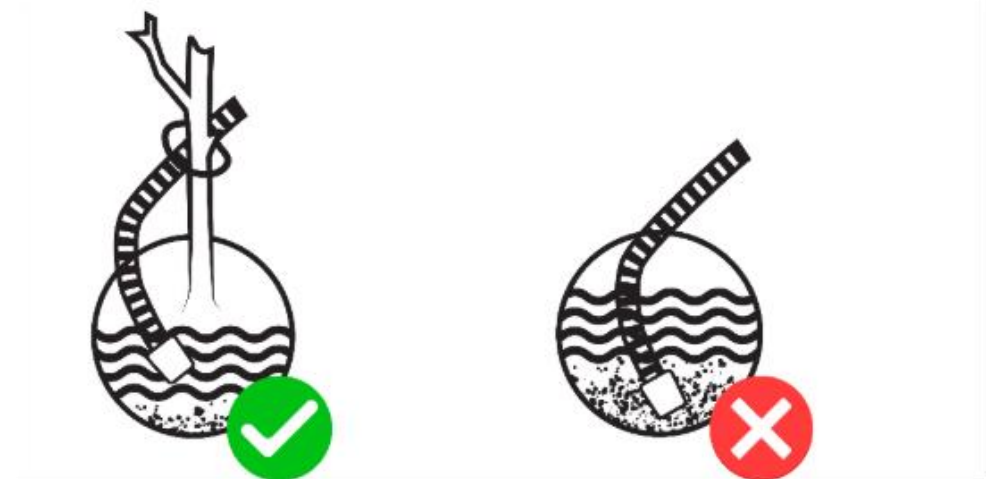
Do not exceed the **suction depth limit** or the pump will be damaged by cavitation:



7m at sea level

6m at 1000 meters of altitude

5m at 2000 meters of altitude



Attach the suction hose to a wooden stick or tree branch so that the foot valve does not touch the bottom of the water source (at least 10 cm from the bottom).

In this way, no debris or sand will be sucked into the pump, which could block the pump mechanism.

How to prepare pressure side:

The outlet is also designed for **1^{1/2}" pipe**.

You can use 1 or 2 m of suction hose to have a flexible outlet (easier to disconnect), then insert inside a PVC pipe (smaller size, 1 ^{1/4}"), which you can then extend to 1 ^{1/2}" or more depending on the pumping distance (for more than 500 m, use 2").



You can also directly use 1 ^{1/2}" HDPE pipe at the outlet.



If you have more than 15m head (altitude difference), use a non-return (check) valve at the outlet to protect



the pump from excessive back pressure.

Use also extra hose clamps.



How to handle a first installation:

The pump needs to be "primed" (the inlet and outlet need to be filled with water), so it would not be running dry.

It is also very important to fill the suction pipe, so the pump will start easier.

Be careful with the loose MC4 connectors (cables) so that you do not accidentally step on them and crush them.



As a general rule, MC4 connectors should not be left on the floor, as they can get dirty with mud and subsequently fail to connect properly.



How to protect pump from future damages:

If you get your water from a muddy lake or river, dig a suction pit next to it so that the pump's internal parts work better in the long run, and you avoid foot valve or pump clogging. Or use a pre-filtration system by placing the foot valve in a bucket or perforated bottle with a mosquito net around it.

